

LPDDR4X高速信号测试记录DDR1

场景:2133Mhz VDEC+NPU压测 测量对象:DDR1

驱动组合:

ADR:DRIVER 40 ODT 240

DQS:DRIVER 40 ODT 60

DQ:DRIVER 40 ODT 60

不做任何调整

测试项	子项1	子项2	子项3
	Singal-Ended Signals	WRITE cycles tests	Clocks Plus Tests
			Clocks Minus Tests
			Clock Tests
			Strobe Plus Tests
			Strobe Minus Tests
			Data Strobe Tests
		READ cycles tests	VOH(AC)
			VOH(DC)
			VOL(AC)
			VOL(DC)
			SRQSeR
			SRQSeF
		Overshoot/ Undershoot (Clock Plus)	Overshoot amplitude (Clock)
			Overshoot area(Clock)
			Undershoot amplitude (Clock)
			Undershoot area (Clock)
		Overshoot/ Undershoot (Clock Minus)	Overshoot amplitude (Clock)
			Overshoot area(Clock)
			Undershoot amplitude (Clock)
			Undershoot area (Clock)
		Overshoot/ Undershoot (Strobe Plus)	Overshoot amplitude (Strobe)
			Overshoot area(Strobe)
			Undershoot amplitude (Strobe)
			Undershoot area (Strobe)

Electrical Tests		Overshoot/ Undershoot (Strobe Minus)	Overshoot amplitude (Strobe)
			Overshoot area(Strobe)
			Undershoot amplitude (Strobe)
			Undershoot area (Strobe)
		Overshoot/ Undershoot (Data)	Overshoot amplitude (Data)
			Overshoot area(Data)
			Undershoot amplitude (Data)
			Undershoot area (Data)
		Overshoot/Undersh (Address, Control)	Overshoot amplitude (Addr)
			Overshoot area (Addr)
			Undershoot amplitude (Addr)
			Undershoot area (Addr)
	Differential Signals (WRITE cycle tests)	Differential AC Input	VIHdiff.CK(AC)
			VIHdiff.CK(DC)
			VILdiff.CK(AC)
			VILdiff.CK(DC)
			Vindiff_CK
			Vindiff_CK/2 High Pulse
			Vindiff_CK/2 Low Pulse
			Vinse_CK (Positive Pulse)
			Vinse_CK (Negative Pulse)
			SRIdiffR for Clock
			SRIdiffF for Clock
		Clock Cross Point Voltage Test	Vix_CK_ratio
		Differential Input Level and Slew Rate For Strobe	VIHdiff_DQS
			VILdiff_DQS
			Vindiff_DQS
			Vindiff_DQS/2 High Pulse
			Vindiff_DQS/2 Low Pulse
			Vinse_DQS (Positive Pulse)
			Vinse_DQS (Negative Pulse)
			SRIdiffR for Strobe
			SRIdiffF for Strobe
		Strobe Cross Point Voltage Test	Vix_DQS_ratio
	Differential Signals	Differential AC	VOHdiff(AC)
			VOLdiff(AC)

	(READ cycle tests)	Output Levels and Slew Rate tests	SRQdiffR
			SRQdiffF
Timing Tests	Timing tests (WRITE cycle tests)	Data Strobe Timing	tDQSS
			tDSS
			tDSH
			tDQSH
			tDQSL
			tWPRE
			tWPST
			tDQSQ
			tQSH
			tQSL
			tQH
	Timing tests (READ cycle tests)	Data Timing	tDQSQ_DBI
			tQSH_DBI
			tQSL_DBI
			tQH_DBI
		Data Strobe Timing	tDQSCK
			tQSH
			tQSL
			tQSH_DBI
			tQSL_DBI
			tRPRE
			tRPST
		Rising Edge Measurements	tCK(abs) Rising Edge Measurements
			tjit(CC) Rising Edge Measurements
			tCK(avg) Rising Edge Measurements
			tjit(per) Rising
			tERR(nper) Rising Edge Measurements
	Timing tests (Clock Timing)	Pulse Measurements	tCK(avg) Rising Edge Measurement
			tCH(abs) Absolute clock HIGH pulse width
			tCL(abs) Absolute clock LOW pulse width
			tCH Average High Measurements
			tjit(duty-high) Jitter Average High Measurements
			tCL Average Low Measurements
			tjit(duty-low) Jitter Average Low Measurements
		tDQ2DQ	

Eye-Diagram for Data and Data Strobe (WRITE)	WRITE cycle tests	tDQS2DQ
		DQ VIH _L (ac)
		tD _{IVW} Margin
		vD _{IVW} Margin
		SRIN _{diVW}
		tD _{IPW}
Eye-Diagram for Data and Data Strobe (READ)	READ cycle tests	tQW _{total} _DBI
		tQW _{total}
Eye-Diagram for Command Address	tC _{IVW} Margin	
	vC _{IVW} Margin	
	CA VIH _L (ac)	
	SRIN _{ciVW}	
	tC _{IPW}	

使用custom模式

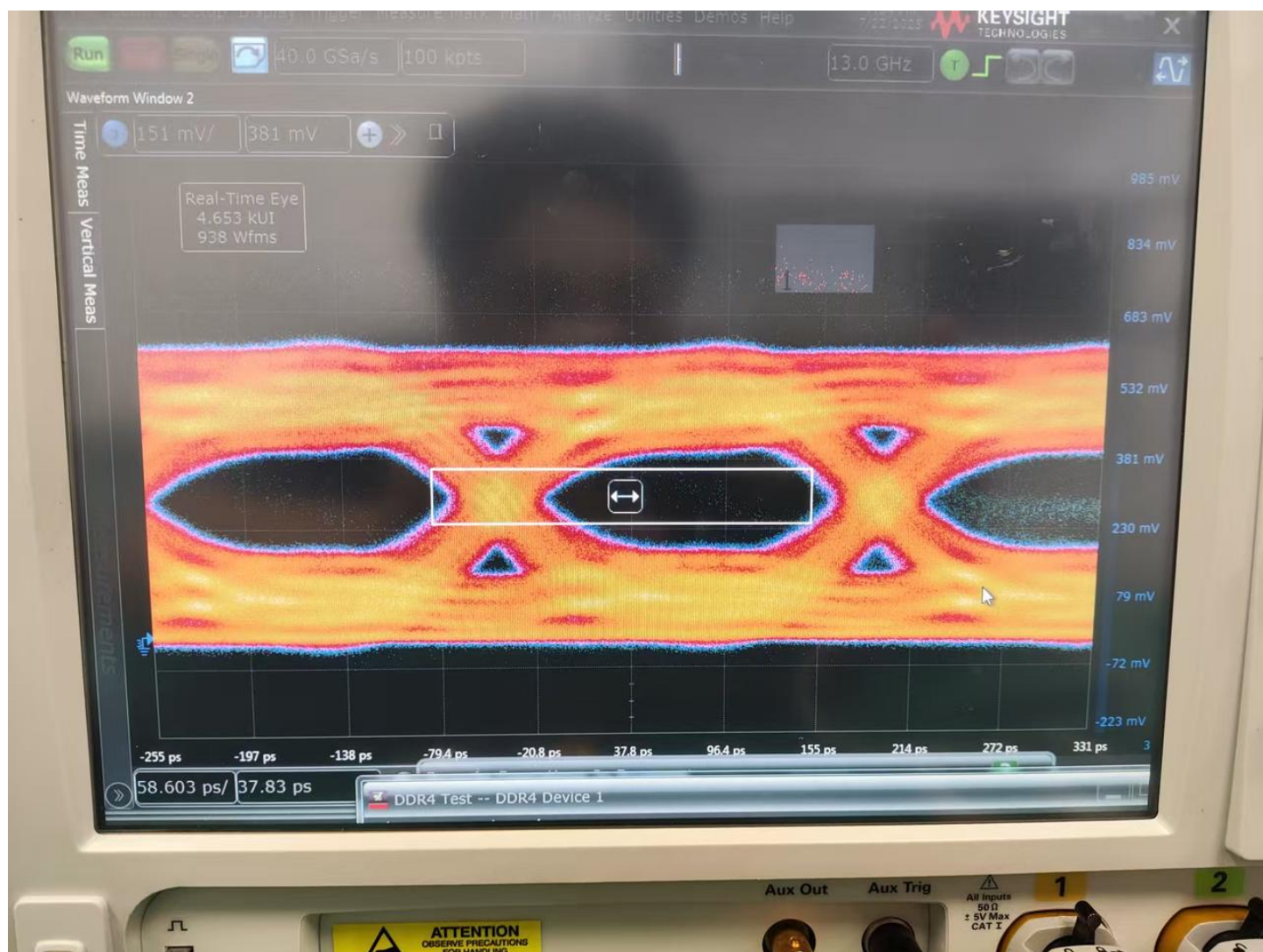
测试项描述请参考文档:

[📄 Keysight D9040DDRC DDR4 Test .pdf](#)

地址线ADDR 3眼图:



数据线DQ0写眼图:



数据线DQ0读眼图:



infiniium

DSA91304A Digital Signal Analyzer

13 GHz
40 GSa/s

MEGA Zoom

File Control Setup Display Trigger Measure/Mark Math Analyze Utilities Demos Help

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7/22/2025



Run Stop Single 40.0 GSa/s 100 kpts

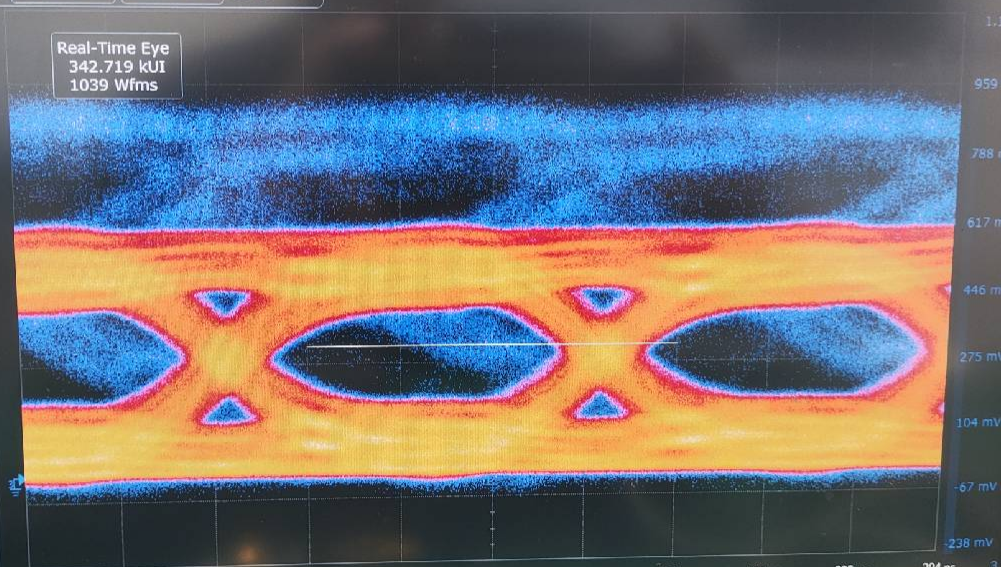
13.0 GHz



Waveform Window 2

Time Meas Vertical Meas 171 mV/ 446 mV

Real-Time Eye
342.719 kUI
1039 Wfms



-292 ps -234 ps -175 ps -116 ps -57.9 ps 700 fs 59.3 ps 118 ps 177 ps 235 ps 294 ps

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Aux Out

Aux Trig

All inputs
50 Ω
+ 8V Max
CAT 1

1

2